

EC602

Computer Network

Combined Module Notes

Modules 1 – 4

Semester VI • 3L:0T:0P • 3 Credits

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Disclaimer / Study Guide Notice:

These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

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Module 1

Overview of Data Communication and Networking

1. Overview of Data Communication and Networking

Data Communication: The exchange of data between two devices via a transmission medium. It requires a *sender*, *receiver*, *message*, *medium*, and *protocol*.

1.1 Components of Data Communication

Component	Description
Message	The information (data) to be communicated — text, numbers, images, audio, video.
Sender	The device that sends the data message (computer, phone, camera).
Receiver	The device that receives the message (computer, printer, phone).
Medium	The physical path over which the message travels (cable, air, fibre).
Protocol	A set of rules governing data communication; ensures sender and receiver agree on format, timing, and error handling.

1.2 Data Representation

Data is represented in different formats depending on the type of information:

Type	Representation
Text	Encoded using bit patterns — ASCII (7-bit, 128 chars), ISO 8859 (8-bit, 256 chars), Unicode (32-bit, universal).
Numbers	Stored directly in binary; not converted to ASCII to save space.
Images	Represented as a matrix of pixels; each pixel encoded in bits (RGB).
Audio	Continuous signal sampled at regular intervals and quantised to binary values.
Video	Sequence of images (frames) displayed at a rate (fps) to create motion.

1.3 Direction of Data Flow

Data Flow Modes

Simplex: 

Half Duplex: 

Full Duplex: 

Mode	Description	Example
Simplex	One direction only; receiver cannot send back.	TV broadcast, keyboard to CPU
Half Duplex	Both directions, but not simultaneously ; one at a time.	Walkie-talkie, CB radio
Full Duplex	Both directions simultaneously ; uses two separate channels or frequencies.	Telephone, mobile call

1.4 Network Criteria

A good network must satisfy three fundamental criteria:

Three Criteria of a Good Network

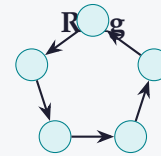
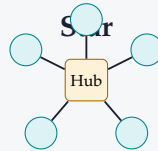
1. **Performance** — measured by transit time (time for a message to travel from sender to receiver) and response time. Affected by number of users, type of medium, hardware, and software.
2. **Reliability** — frequency of failure, recovery time after failure, robustness in catastrophe.
3. **Security** — protecting data from unauthorised access, damage, and implementing recovery policies.

2. Physical Structure: Topology and Categories

2.1 Network Topologies

Topology: The geometric arrangement of devices (nodes) and links in a network. It defines how devices are physically or logically connected.

Common Network Topologies



Topology	Description	Advantage / Disadvantage
Bus	All devices share a single backbone cable. Terminators at both ends.	Simple, cheap / Single point of failure; collisions.
Star	All devices connect to a central hub/switch.	Easy fault isolation / Hub failure = network down.
Ring	Each device connects to exactly two others forming a closed loop.	Equal access / One break disrupts entire ring.
Mesh	Every device connects to every other device (full mesh) or some (partial mesh).	Highly reliable / Expensive, complex wiring.
Tree (Hybrid)	Hierarchical combination of star and bus topologies.	Scalable / Backbone failure affects subtrees.

2.2 Categories of Networks

Category	Coverage	Description & Examples
LAN	Up to a few km	Local Area Network; single building or campus. Ethernet, Wi-Fi. High speed (100 Mbps–10 Gbps).
MAN	City-wide (5–50 km)	Metropolitan Area Network; connects multiple LANs in a city. Cable TV networks, WiMAX.
WAN	Country / Global	Wide Area Network; connects cities, countries. Internet, leased lines, satellite links.

3. Internet: History and Protocols

Internet: A global network of networks interconnected using the TCP/IP protocol suite. It evolved from ARPANET (1969), a US Department of Defense project for fault-tolerant communication.

Brief History:

- **1969** — ARPANET launched; first 4-node network (UCLA, Stanford, UCSB, Utah).
- **1974** — TCP/IP protocol defined by Cerf and Kahn.
- **1983** — ARPANET officially adopts TCP/IP; modern Internet born.

- **1991** — World Wide Web (WWW) introduced by Tim Berners-Lee at CERN.
- **1993** — Mosaic browser; Internet becomes publicly accessible.

Protocols and Standards: A **protocol** is a set of rules governing communication. A **standard** is a protocol agreed upon by industry bodies. Key bodies: **ISO, IEEE, IETF, ITU-T, ANSI**.

4. Reference Models

4.1 OSI Reference Model

OSI Model (Open Systems Interconnection): A 7-layer conceptual framework developed by ISO (1984) to standardise communication between different systems. Each layer has specific functions and communicates with adjacent layers via interfaces.

#	Layer	Function & PDU
7	Application	User interface; provides network services to applications. PDU: <i>Data</i> . Protocols: HTTP, FTP, SMTP, DNS.
6	Presentation	Translation, encryption, compression of data. PDU: <i>Data</i> .
5	Session	Establishes, manages, and terminates sessions between applications. PDU: <i>Data</i> .
4	Transport	End-to-end delivery, segmentation, flow control, error control. PDU: <i>Segment</i> . Protocols: TCP, UDP.
3	Network	Logical addressing (IP), routing, path determination. PDU: <i>Packet</i> . Protocols: IP, ICMP, RIP, OSPF.
2	Data Link	Framing, physical addressing (MAC), error detection, flow control. PDU: <i>Frame</i> . Protocols: Ethernet, HDLC.
1	Physical	Transmission of raw bits over medium; defines electrical/optical signals. PDU: <i>Bits</i> .

4.2 TCP/IP Reference Model

TCP/IP Model: A 4-layer practical model that forms the basis of the Internet. Developed by the DARPA. Less abstract than OSI; combines OSI layers 5, 6, 7 into Application and layers 1, 2 into Network Access.

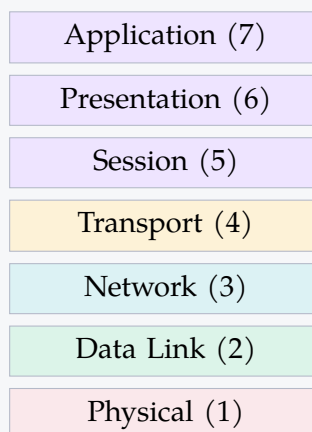
TCP/IP Layer	OSI Equivalent	Protocols
Application	Layers 5, 6, 7	HTTP, FTP, SMTP, DNS, SNMP, Telnet
Transport	Layer 4	TCP, UDP
Internet	Layer 3	IP (IPv4, IPv6), ICMP, ARP, RARP
Network Access	Layers 1, 2	Ethernet, Wi-Fi (802.11), Token Ring, Frame Relay

4.3 OSI vs. TCP/IP — Comparison

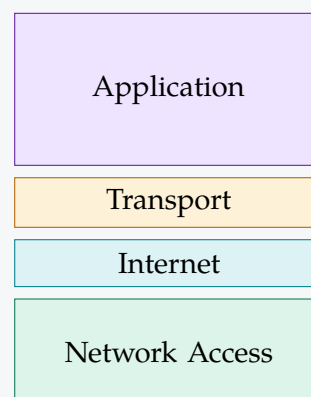
Parameter	OSI Model	TCP/IP Model
Layers	7 layers	4 layers
Developed by	ISO (1984)	DARPA (1970s)
Nature	Conceptual / theoretical	Practical / implemented
Transport layer	Connection & connectionless	Connectionless (UDP) + Connection (TCP)
Session/Presentation	Separate layers (5 & 6)	Merged into Application layer
Protocol dependency	Protocol-independent model	Protocol-specific (TCP/IP suite)
Usage	Reference / teaching tool	Actual Internet implementation

OSI vs. TCP/IP Layer Mapping

OSI (7 layers)



TCP/IP (4 layers)



5. Physical Level

5.1 Analog vs. Digital Data and Signals

Aspect	Analog	Digital
Data	Continuous values (e.g. voice, temperature)	Discrete values (0 and 1)
Signal	Continuous waveform (sine wave)	Square wave / pulse train
Noise immunity	Low — noise degrades signal	High — regenerated at repeaters
Bandwidth	Lower bandwidth needed	Higher bandwidth required
Transmission	Analog transmission (amplifiers)	Digital transmission (repeaters)

5.2 Transmission Media

Transmission Medium: The physical path between sender and receiver. Classified as **guided** (wired) or **unguided** (wireless).

Guided Media:

- **Twisted Pair Cable:** Two insulated copper wires twisted together. Reduces electromagnetic interference. Types: UTP (Unshielded) and STP (Shielded). Used in telephone lines, Ethernet (Cat5e, Cat6).
- **Coaxial Cable:** Central conductor surrounded by insulation, metallic shield, and outer jacket. Higher bandwidth than twisted pair. Used in cable TV, older Ethernet (10Base2, 10Base5).
- **Fibre Optic Cable:** Transmits data as light pulses through glass/plastic core. Extremely high bandwidth, immune to EMI, low attenuation. Used in backbone networks, long-distance links.

Unguided Media (Wireless):

- **Radio Waves:** Omnidirectional; penetrate walls. Used in AM/FM radio, Wi-Fi, Bluetooth.
- **Microwaves:** Directional; line-of-sight required. Used in satellite communication, cellular networks.
- **Infrared:** Short range; cannot penetrate walls. Used in TV remotes, IrDA.

5.3 Circuit Switching

Circuit Switching: A dedicated communication path is established between sender and receiver for the entire duration of the session. Resources are reserved end-to-end. Used in traditional telephone networks (PSTN).

Phases of Circuit Switching:

1. **Circuit Establishment:** A dedicated path is set up through intermediate switches.
2. **Data Transfer:** Data flows continuously along the reserved path.
3. **Circuit Disconnect:** Path is released and resources freed after communication ends.

Space Division Switch

Space Division Switching

Each connection uses a **physically separate path** through the switch matrix. A crosspoint (transistor/relay) is activated at the intersection of input and output lines.

Crosspoint matrix: For n inputs and n outputs, requires n^2 crosspoints.

Advantage: Simple, simultaneous connections. **Disadvantage:** Expensive for large n .

Time Division Switch (TDM Bus)

Time Division Switching

All connections share the **same physical medium** but are separated in **time slots**. A Time Slot Interchange (TSI) unit reads input slots and writes them to output slots.

TDM Bus: A high-speed shared bus; each input is assigned a time slot. The TSI reorders slots to route data to correct outputs.

Advantage: Efficient use of medium. **Disadvantage:** Requires synchronisation.

5.4 Telephone Network

The Public Switched Telephone Network (PSTN) uses circuit switching. It consists of:

- **Local loops:** Twisted pair from subscriber to local exchange (last mile).
- **Trunks:** High-capacity links (fibre/microwave) between exchanges.
- **Switching offices:** Local, tandem, and toll offices that route calls.

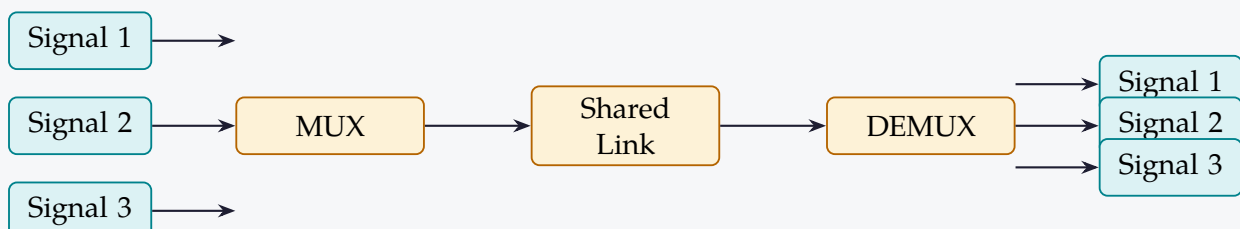
Telephone Network Structure



6. Multiplexing: FDM, TDM, and WDM

Multiplexing: A technique that allows multiple signals to share a single transmission medium simultaneously. A **multiplexer (MUX)** combines signals at the sender; a **demultiplexer (DEMUX)** separates them at the receiver.

General Multiplexing Concept



6.1 Frequency Division Multiplexing (FDM)

FDM (Frequency Division Multiplexing): The total bandwidth of the medium is divided into non-overlapping **frequency bands (sub-channels)**. Each signal is modulated onto a different carrier frequency and transmitted simultaneously. Used for **analog signals**.

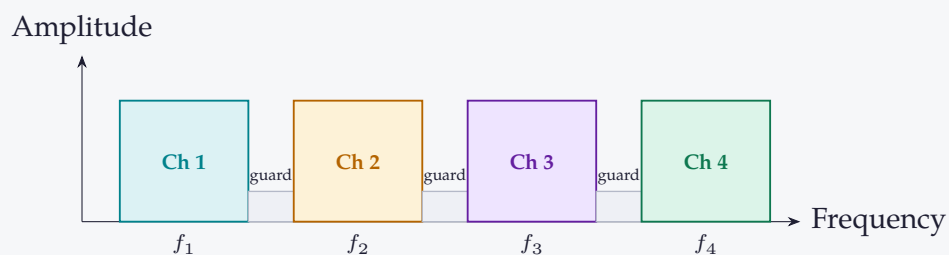
Working:

1. Each input signal is modulated onto a unique carrier frequency $f_1, f_2, \dots, f_n$.
2. Guard bands (unused frequency gaps) separate adjacent channels to prevent interference.
3. All modulated signals are combined and transmitted over the shared medium.
4. At the receiver, bandpass filters separate each channel; demodulation recovers the original signal.

FDM Key Formulas

Channel bandwidth	$B_{ch} = f_{upper} - f_{lower}$
Total bandwidth	$B_{total} = n \times B_{ch} + (n - 1) \times B_{guard}$
Guard band	Unused frequency gap between adjacent channels to prevent inter-channel interference
Carrier frequencies	$f_1 < f_2 < \dots < f_n$, each separated by at least $B_{ch} + B_{guard}$

FDM — Frequency Spectrum Allocation



Aspect	Details
Signal type	Analog signals
Separation method	Different carrier frequencies; guard bands prevent overlap
Simultaneous?	Yes — all channels transmit at the same time
Applications	AM/FM radio, cable TV, ADSL, first-generation cellular (1G)
Advantage	Simple; no synchronisation needed; all channels always active
Disadvantage	Guard bands waste bandwidth; susceptible to intermodulation noise

6.2 Time Division Multiplexing (TDM)

TDM (Time Division Multiplexing): The time on the shared channel is divided into fixed **time slots**. Each input signal is assigned a slot in a repeating **frame**. Used for **digital signals**.

Two types of TDM:

Synchronous TDM

Each input is assigned a **fixed time slot** in every frame, regardless of whether it has data to send. Slots are pre-allocated.

Frame duration $T_f = n \times T_s$ (n = number of channels, T_s = slot duration)

Output bit rate $R_{out} = n \times R_{in}$ (output must be n times faster than each input)

Frame efficiency Can be low — empty slots wasted if a channel has no data

Statistical (Asynchronous) TDM

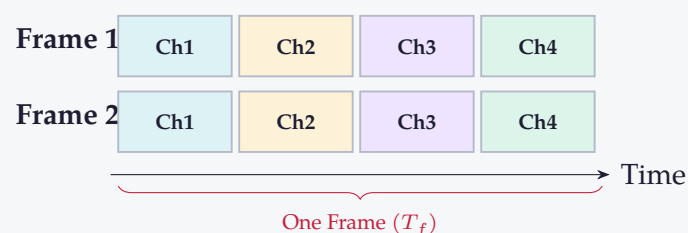
Slots are allocated **dynamically** — only to channels that have data. Each slot carries an **address header** to identify the source.

Efficiency Higher than synchronous TDM; no wasted slots

Overhead Address field added to each slot increases overhead

Output bit rate Less than $n \times R_{in}$ (statistical multiplexing gain)

Synchronous TDM — Frame Structure (4 channels)



Aspect	Synchronous TDM	Statistical TDM
Slot allocation	Fixed; pre-assigned per channel	Dynamic; only active channels get slots
Address header	Not needed (position identifies channel)	Required (slot carries source address)
Efficiency	Low if channels idle	High; no wasted slots
Output rate	$n \times R_{in}$ (fixed)	Less than $n \times R_{in}$
Example	T1/E1 lines, SONET	ATM, packet-switched networks

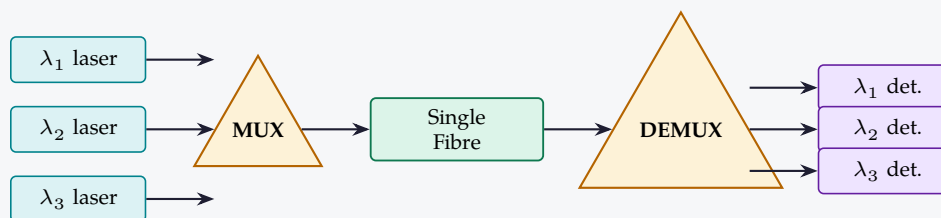
6.3 Wavelength Division Multiplexing (WDM)

WDM (Wavelength Division Multiplexing): Multiple optical signals are transmitted simultaneously over a single **fibre optic cable**, each on a different **wavelength (colour) of light**. Conceptually identical to FDM but applied to optical frequencies. Used in high-capacity fibre networks.

Working:

1. Each data stream modulates a laser of a distinct wavelength $\lambda_1, \lambda_2, \dots, \lambda_n$.
2. A **prism or optical multiplexer** combines all wavelengths onto one fibre.
3. The combined beam travels through the fibre.
4. At the receiver, a **prism or optical demultiplexer** separates the wavelengths.
5. Each wavelength is detected by a photodetector and demodulated.

WDM — Wavelength Multiplexing on Fibre



Types of WDM:

- **CWDM (Coarse WDM):** Fewer channels (up to 18), wider wavelength spacing (20 nm). Lower cost, shorter distances.
- **DWDM (Dense WDM):** Many channels (40–160+), very narrow spacing (0.8 nm or less). Used in long-haul backbone networks. Capacity up to several Tbps on one fibre.

Aspect	Details
Medium	Fibre optic cable only
Separation method	Different wavelengths (colours) of light
Analogy	FDM applied to optical domain
Capacity	DWDM: 40–160+ channels; Tbps aggregate
Applications	Internet backbone, long-haul telecom, submarine cables
Advantage	Enormous capacity; each wavelength is independent
Disadvantage	Expensive equipment; requires precise laser wavelengths

6.4 FDM vs. TDM vs. WDM — Comparison

Parameter	FDM	TDM	WDM
Full form	Frequency Division Multiplexing	Time Division Multiplexing	Wavelength Division Multiplexing
Signal type	Analog	Digital	Optical (light)
Sharing basis	Frequency bands	Time slots	Wavelengths of light
Medium	Coaxial, twisted pair, air	Any digital medium	Fibre optic only
Synchronisation	Not required	Required (sync TDM)	Not required
Guard bands	Required	Not needed	Not needed (wavelength spacing)
Applications	Radio, cable TV, ADSL	T1/E1, SONET, GSM	Internet backbone, DWDM
Capacity	Moderate	High	Very high (Tbps)

Quick Revision — Module 1

Data Communication, Networking Overview & Physical Level | EC602

Key Facts at a Glance

OSI Layers (top to bottom)	Application, Presentation, Session, Transport, Network, Data Link, Physical
TCP/IP Layers	Application, Transport, Internet, Network Access
LAN coverage	Up to a few km (building/campus)
MAN coverage	City-wide (5–50 km)
WAN coverage	Country / global
Simplex	One direction only
Half Duplex	Both directions, not simultaneously
Full Duplex	Both directions simultaneously
Circuit switching phases	Establish → Transfer → Disconnect
Space division crosspoints	n^2 for $n \times n$
TDM output bit rate	$R_{out} = n \times R_{in}$ (synchronous TDM)
TDM frame duration	$T_f = n \times T_s$
FDM total bandwidth	$B_{total} = n \times B_{ch} + (n - 1) \times B_{guard}$
WDM	FDM applied to optical domain; uses wavelengths of light switch
ASCII	7-bit encoding, 128 characters
Unicode	32-bit encoding, universal character set

One-Line Definitions

- **Protocol:** Set of rules governing communication between devices.
- **Topology:** Geometric arrangement of devices and links in a network.
- **LAN:** Local Area Network; high-speed network within a building or campus.
- **WAN:** Wide Area Network; spans cities, countries, or the globe.
- **OSI Model:** 7-layer ISO standard for network communication.
- **TCP/IP Model:** 4-layer practical model underlying the Internet.
- **Circuit Switching:** Dedicated path reserved for entire session duration.

- **TDM:** Time Division Multiplexing; shares medium by assigning time slots to each channel.
- **FDM:** Frequency Division Multiplexing; divides bandwidth into frequency bands for analog signals.
- **WDM:** Wavelength Division Multiplexing; uses different light wavelengths on a single fibre optic cable.
- **Statistical TDM:** Dynamic slot allocation; only active channels get slots, improving efficiency.
- **DWDM:** Dense WDM; 40–160+ channels with narrow wavelength spacing; used in backbone networks.
- **Guided Media:** Physical wired medium (twisted pair, coaxial, fibre optic).
- **Unguided Media:** Wireless medium (radio, microwave, infrared).
- **Full Duplex:** Simultaneous two-way communication.
- **ASCII:** American Standard Code for Information Interchange; 7-bit text encoding.

Frequently Asked / Viva Questions

Q1. What are the five components of data communication?

Sender, Receiver, Message, Medium, and Protocol.

Q2. Difference between simplex, half duplex, and full duplex?

Simplex: one direction only. Half duplex: both directions but not simultaneously. Full duplex: both directions simultaneously.

Q3. How many layers does the OSI model have? Name them.

7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application (bottom to top).

Q4. What is the PDU at the Network layer?

Packet.

Q5. What is the PDU at the Data Link layer?

Frame.

Q6. How does TCP/IP differ from OSI?

TCP/IP has 4 layers vs. OSI's 7; TCP/IP is practical and protocol-specific; OSI is a conceptual reference model.

Q7. What is circuit switching?

A dedicated communication path is established and reserved for the entire session. Used in PSTN.

Frequently Asked / Viva Questions (contd.)**Q8. What is the difference between space division and time division switching?**

Space division uses separate physical paths (crosspoint matrix); time division shares one medium using time slots (TSI).

Q9. What is the advantage of fibre optic over copper?

Higher bandwidth, lower attenuation, immune to EMI, more secure.

Q10. What is a MAN?

Metropolitan Area Network; covers a city (5–50 km); e.g. cable TV network, WiMAX.

Q11. What does ASCII stand for?

American Standard Code for Information Interchange; 7-bit encoding for 128 characters.

Q12. Why is full duplex preferred over half duplex?

Full duplex allows simultaneous two-way communication, doubling effective throughput compared to half duplex.

Q13. What is multiplexing and why is it used?

Multiplexing allows multiple signals to share a single transmission medium, reducing cost and making efficient use of bandwidth.

Q14. What is the difference between FDM and TDM?

FDM divides the bandwidth into frequency bands for analog signals; TDM divides time into slots for digital signals. FDM channels transmit simultaneously; TDM channels take turns.

Q15. What is WDM and where is it used?

WDM transmits multiple optical signals on one fibre using different wavelengths of light. Used in Internet backbone and long-haul telecom (DWDM supports 40–160+ channels).

Q16. What is the difference between synchronous and statistical TDM?

Synchronous TDM assigns fixed slots to each channel regardless of activity (wastes slots if idle). Statistical TDM allocates slots dynamically only to active channels, improving efficiency but requiring address headers.

OSI Layer Summary Table

#	Layer	Key Function	PDU
7	Application	User-facing services (HTTP, FTP, DNS)	Data
6	Presentation	Encryption, compression, translation	Data
5	Session	Session management, synchronisation	Data
4	Transport	End-to-end delivery, TCP/UDP	Segment
3	Network	Routing, IP addressing	Packet
2	Data Link	Framing, MAC, error detection	Frame
1	Physical	Bit transmission over medium	Bits

Module 2

Data Link Layer

1. Data Link Layer

Data Link Layer (Layer 2): Responsible for *node-to-node* delivery of frames. It handles framing, physical addressing (MAC), error detection/correction, and flow control between adjacent nodes.

1.1 Types of Errors

Error Type	Description
Single-bit error	Only one bit in the data unit is changed (0 to 1 or 1 to 0). Rare in serial transmission.
Burst error	Two or more consecutive bits are corrupted. More common; caused by noise bursts. Length = from first to last corrupted bit.

1.2 Framing

Framing: The process of dividing a stream of bits from the network layer into manageable data units called **frames**. Allows the receiver to identify the start and end of each frame.

Character (Byte) Stuffing

Used when frames contain text data. A special **flag byte** (e.g. `FLAG`) marks frame boundaries. If the flag byte appears in the data, an **escape byte** (`ESC`) is inserted before it.

- If data contains `FLAG`: insert `ESC` before it.
- If data contains `ESC`: insert another `ESC` before it.
- Receiver removes the stuffed `ESC` bytes.

Bit Stuffing

Used in bit-oriented protocols (e.g. HDLC). Flag pattern is `01111110`. To prevent this pattern from appearing in data:

- **Sender:** After every 5 consecutive 1s in data, insert a 0.
- **Receiver:** After every 5 consecutive 1s, if next bit is 0, remove it; if 1, it is a flag.

1.3 Error Detection and Correction

Parity Check

Parity Check

A single parity bit is appended to make the total number of 1s **even** (even parity) or **odd** (odd parity).

Limitation: Detects only *odd* number of bit errors; cannot detect even-number errors or correct errors.

Cyclic Redundancy Check (CRC)

CRC — Most Widely Used Error Detection

Concept: Treat the data as a binary polynomial. Divide by a **generator polynomial** $G(x)$. Append the remainder (CRC bits) to the data.

Sender:

1. Append r zeros to data ($r = \text{degree of } G(x)$).
2. Divide augmented data by $G(x)$ using modulo-2 division (XOR).
3. Replace appended zeros with remainder $R(x)$ — this is the CRC.

Receiver: Divide received frame by $G(x)$. If remainder = 0, no error; else error detected.

Common generators: CRC-8 ($x^8 + x^2 + x + 1$), CRC-16, CRC-32 (Ethernet).

Hamming Code (Error Correction)

Hamming Code

Adds **redundant parity bits** at positions that are powers of 2 ($p_1, p_2, p_4, p_8, \dots$). Each parity bit covers specific data bits.

Minimum redundant bits: For m data bits, need r parity bits such that $2^r \geq m + r + 1$.

Error correction: Receiver recalculates parity bits; the binary value of failing parity positions gives the **error position** (syndrome). Flip that bit to correct.

Capability: Detects and corrects **single-bit errors**; detects (not corrects) double-bit errors.

1.4 Flow Control

Flow Control: Mechanism to prevent a fast sender from overwhelming a slow receiver. The receiver controls the rate at which the sender transmits data.

Method	Description
Stop-and-Wait	Sender sends one frame, waits for ACK before sending next. Simple but inefficient (low throughput).
Sliding Window	Sender can transmit multiple frames (window size W) before needing ACK. Improves efficiency.

1.5 ARQ Protocols (Automatic Repeat reQuest)

ARQ: Error control mechanism combining error detection with retransmission. Uses ACK (acknowledgement) and NAK (negative acknowledgement) to ensure reliable delivery.

Stop-and-Wait ARQ

Stop-and-Wait ARQ

Sender transmits one frame and starts a timer. Waits for ACK.

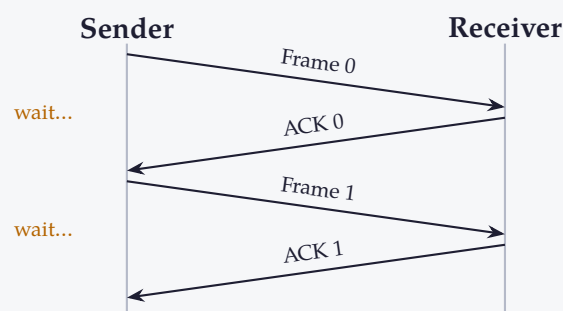
If ACK received: Send next frame.

If NAK or timeout: Retransmit the same frame.

Efficiency: $\eta = \frac{1}{1 + 2a}$ where $a = \frac{T_p}{T_f}$ (propagation delay / frame transmission time).

Drawback: Very low efficiency for large a (long links or short frames).

Stop-and-Wait ARQ — Message Flow



Go-Back-N ARQ (GBN)

Go-Back-N ARQ

Sender window size $W_s = 2^n - 1$ (for n -bit sequence numbers). Receiver window $W_r = 1$.

On error: Receiver discards the erroneous frame and all subsequent frames. Sender retransmits from the erroneous frame onwards.

$$\text{Efficiency: } \eta = \begin{cases} \frac{1}{1+2a} & \text{if } W \geq 1+2a \\ \frac{W}{1+2a} & \text{if } W < 1+2a \end{cases}$$

Drawback: Wastes bandwidth retransmitting correctly received frames.

Selective Repeat ARQ (SR)

Selective Repeat ARQ

Sender window $W_s = 2^{n-1}$. Receiver window $W_r = 2^{n-1}$.

On error: Only the **erroneous frame** is retransmitted. Receiver buffers out-of-order frames.

$$\text{Efficiency: } \eta = \begin{cases} 1 & \text{if } W \geq 1+2a \\ \frac{W}{1+2a} & \text{if } W < 1+2a \end{cases}$$

Advantage: More efficient than GBN; **Disadvantage:** Requires larger receiver buffer.

Parameter	Stop-and-Wait	Go-Back-N	Selective Repeat
Sender window	1	$2^n - 1$	2^{n-1}
Receiver window	1	1	2^{n-1}
On error	Retransmit 1	Retransmit from error	Retransmit only error
Efficiency	Lowest	Medium	Highest
Buffer needed	Minimal	Small	Large

1.6 HDLC (High-level Data Link Control)

HDLC: A bit-oriented, synchronous data link protocol developed by ISO. Uses bit stuffing for framing. Supports point-to-point and multipoint links.

HDLC Frame Structure:

Flag	Address	Control	Data	FCS + Flag
01111110	8 bits	8 or 16 bits	Variable	16/32 bits + 01111110

HDLC Frame Types:

- **I-frame (Information):** Carries user data; also piggybacks ACK.
- **S-frame (Supervisory):** Flow and error control (ACK, NAK, RNR).
- **U-frame (Unnumbered):** Link management (setup, disconnect).

2. Medium Access Sub-layer

Medium Access Control (MAC): Sub-layer of the Data Link layer that controls how devices on a shared medium gain access to transmit. Prevents collisions and manages channel allocation.

2.1 Point-to-Point Protocol (PPP)

PPP: A byte-oriented protocol for direct serial links (e.g. dial-up, DSL). Provides framing, authentication, and network layer protocol negotiation.

PPP Components:

- **LCP (Link Control Protocol):** Establishes, configures, and tests the data link connection.
- **NCP (Network Control Protocol):** Configures network layer protocols (e.g. IPCP for IP). One NCP per network layer protocol.

PPP Frame: Flag (01111110) | Address (FF) | Control (03) | Protocol | Data | FCS | Flag

2.2 Token Ring

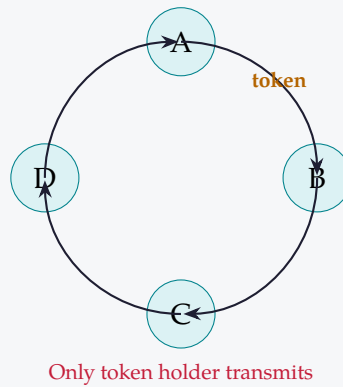
Token Ring (IEEE 802.4/802.5): A deterministic MAC protocol where a special bit pattern called a **token** circulates around a ring. Only the station holding the token can transmit.

Operation:

1. A free token circulates continuously around the ring.
2. A station wishing to transmit captures the token, changes it to a *busy token*, and appends its frame.
3. The frame travels around the ring; the destination copies the data.
4. The originating station removes the frame and releases a new free token.

Advantage: Deterministic, no collisions, fair access. **Disadvantage:** Token loss/corruption disrupts network.

Token Ring Operation



2.3 Multiple Access Protocols

Pure ALOHA

Pure ALOHA

Stations transmit whenever they have data. If collision occurs, wait a **random backoff time** and retransmit.

Throughput: $S = G \cdot e^{-2G}$, maximum $S_{max} = \frac{1}{2e} \approx 18.4\%$ at $G = 0.5$.

Vulnerable period: $2T_f$ (two frame times).

Slotted ALOHA

Slotted ALOHA

Time is divided into slots of size T_f . Stations can only transmit at the **beginning of a slot**.

Throughput: $S = G \cdot e^{-G}$, maximum $S_{max} = \frac{1}{e} \approx 36.8\%$ at $G = 1$.

Vulnerable period: T_f (one frame time) — half that of Pure ALOHA.

CSMA (Carrier Sense Multiple Access)

CSMA: Before transmitting, a station **senses the channel**. If busy, it waits; if idle, it transmits. Reduces (but does not eliminate) collisions due to propagation delay.

CSMA Variants:

Variant	Behaviour on Busy Channel
1-persistent	Wait until idle, then transmit immediately with probability 1. High collision risk.
Non-persistent	Wait a random time, then sense again. Lower collision risk but higher delay.
p-persistent	When idle, transmit with probability p , defer with probability $1 - p$. Compromise.

CSMA/CD (Collision Detection) — Ethernet

CSMA/CD

Used in wired Ethernet (IEEE 802.3). Station senses channel; if idle, transmits and **monitors for collision simultaneously**.

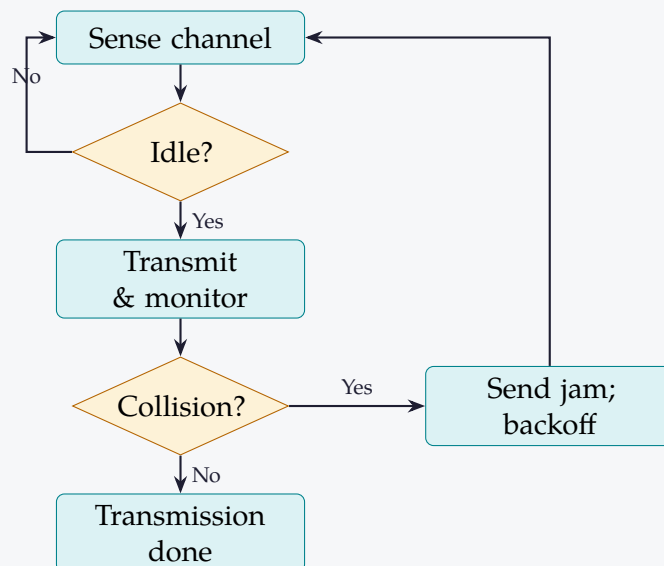
On collision: Transmit jam signal, stop transmission, wait **binary exponential backoff** time, retry.

Backoff time: $t = r \times T_{slot}$ where r is random in $[0, 2^k - 1]$, k = attempt number (max 10).

Minimum frame size: $L_{min} = 2 \times T_p \times B$ (to detect collision before frame ends).

Not used in wireless (cannot detect collision while transmitting).

CSMA/CD Flowchart



CSMA/CA (Collision Avoidance) — Wi-Fi

CSMA/CA

Used in wireless networks (IEEE 802.11). Collision **avoidance** instead of detection (cannot detect collision in wireless).

Mechanism:

1. Sense channel; if idle for DIFS time, transmit.
2. If busy, wait for channel to be idle, then wait a random **backoff** time.
3. Transmit; receiver sends ACK after SIFS time.
4. Optional: use **RTS/CTS** (Request to Send / Clear to Send) to reserve channel.

2.4 Traditional and Fast Ethernet

Parameter	Traditional Ethernet (10 Mbps)	Fast Ethernet (100 Mbps)
Standard	IEEE 802.3	IEEE 802.3u
Speed	10 Mbps	100 Mbps
MAC protocol	CSMA/CD	CSMA/CD
Frame format	Same as 802.3	Same as 802.3
Media	Coaxial, UTP, fibre	UTP Cat5, fibre
Variants	10Base2, 10Base5, 10BaseT	100BaseTX, 100BaseFX, 100BaseT4

Quick Revision — Module 2

Data Link Layer & Medium Access Sub-layer | EC602

Key Formulas at a Glance

Stop-and-Wait efficiency	$\eta = \frac{1}{1 + 2a}, \quad a = T_p/T_f$
GBN efficiency	$\eta = \frac{W}{1 + 2a}$ if $W < 1 + 2a$, else 1
SR efficiency	$\eta = \frac{W}{1 + 2a}$ if $W < 1 + 2a$, else 1
GBN sender window	$W_s = 2^n - 1$
SR sender/receiver window	$W_s = W_r = 2^{n-1}$
Pure ALOHA max throughput	$S_{max} = 1/(2e) \approx 18.4\%$
Slotted ALOHA max throughput	$S_{max} = 1/e \approx 36.8\%$
Hamming redundant bits	$2^r \geq m + r + 1$

One-Line Definitions

- **Framing:** Dividing bit stream into frames with identifiable start/end boundaries.
- **Bit Stuffing:** Inserting a 0 after five consecutive 1s to prevent flag pattern in data.
- **CRC:** Error detection using polynomial division; remainder appended as check bits.
- **Hamming Code:** Error-correcting code that adds parity bits at power-of-2 positions.
- **ARQ:** Automatic Repeat reQuest; error control using ACK/NAK and retransmission.
- **Go-Back-N:** On error, retransmit erroneous frame and all subsequent frames.
- **Selective Repeat:** On error, retransmit only the erroneous frame.
- **CSMA/CD:** Sense before transmit; detect collision and use backoff (Ethernet).
- **CSMA/CA:** Sense before transmit; avoid collision using backoff and ACK (Wi-Fi).
- **Token Ring:** Only the station holding the token can transmit; deterministic access.
- **PPP:** Point-to-Point Protocol; uses LCP for link setup and NCP for network config.
- **HDLC:** Bit-oriented DLL protocol; uses bit stuffing; I/S/U frame types.

Frequently Asked / Viva Questions**Q1. What is the difference between burst error and single-bit error?**

Single-bit: only one bit corrupted. Burst error: two or more consecutive bits corrupted; length measured from first to last corrupted bit.

Q2. How does bit stuffing work in HDLC?

After every 5 consecutive 1s in data, sender inserts a 0. Receiver removes the stuffed 0 after 5 consecutive 1s.

Q3. What is the maximum throughput of Pure ALOHA and Slotted ALOHA?

Pure ALOHA: $\approx 18.4\%$. Slotted ALOHA: $\approx 36.8\%$.

Q4. Why is Slotted ALOHA more efficient than Pure ALOHA?

Vulnerable period is halved (T_f vs. $2T_f$) because transmission is restricted to slot boundaries.

Q5. What is the sender window size in Go-Back-N for 3-bit sequence numbers?

$$W_s = 2^3 - 1 = 7.$$

Q6. Why can't CSMA/CD be used in wireless networks?

A wireless station cannot detect a collision while transmitting because its own signal overwhelms any incoming signal.

Q7. What are LCP and NCP in PPP?

LCP establishes and configures the link. NCP configures the network layer protocol (e.g. IPCP for IP).

Q8. What is the minimum frame size requirement in CSMA/CD?

Frame transmission time must be $\geq 2 \times$ propagation delay so the sender is still transmitting when the collision signal arrives.

Q9. What are the three frame types in HDLC?

I-frame (data), S-frame (supervisory/flow control), U-frame (link management).

Q10. What is binary exponential backoff?

After k -th collision, station waits a random time in $[0, 2^k - 1]$ slot times before retrying. Reduces repeated collisions.

Q11. Difference between GBN and Selective Repeat?

GBN retransmits erroneous frame and all subsequent frames; SR retransmits only the erroneous frame. SR needs larger receiver buffer.

Q12. What is the role of the MAC sub-layer?

Controls access to the shared transmission medium; prevents and manages collisions; provides physical addressing (MAC address).

Module 3

Network Layer

1. Network Layer

Network Layer (Layer 3): Responsible for *source-to-destination* delivery of packets across multiple networks. Handles logical addressing (IP), routing, and internetworking.

1.1 Internetworking Devices

Device	OSI Layer	Function
Repeater	Layer 1	Regenerates and amplifies signals; extends cable length. No filtering.
Hub	Layer 1	Multi-port repeater; broadcasts to all ports. No intelligence.
Bridge	Layer 2	Connects two LAN segments; filters frames using MAC address table.
Switch	Layer 2	Multi-port bridge; creates dedicated paths between ports; reduces collisions.
Router	Layer 3	Routes packets between different networks using IP addresses and routing tables.
Gateway	Layer 4–7	Protocol converter; connects networks using different protocols (e.g. LAN to mainframe).

1.2 IP Addressing

IP Address: A 32-bit (IPv4) logical address assigned to each device on a network. Written in dotted-decimal notation (e.g. 192.168.1.1). Consists of **Network ID** and **Host ID**.

IPv4 Address Classes:

Class	Range (1st octet)	Default Mask	Use
A	1 – 126	255.0.0.0	Large networks; 126 networks, ~16M hosts each
B	128 – 191	255.255.0.0	Medium networks; 16384 networks, 65534 hosts each
C	192 – 223	255.255.255.0	Small networks; ~2M networks, 254 hosts each
D	224 – 239	—	Multicast
E	240 – 255	—	Reserved / experimental

1.3 Subnetting

Subnetting

Concept: Dividing a large network into smaller sub-networks (subnets) by borrowing bits from the Host ID portion.

Subnet Mask: A 32-bit number with 1s for network+subnet bits and 0s for host bits.

Key formulas:

- Number of subnets = 2^s (where s = borrowed bits)
- Hosts per subnet = $2^h - 2$ (where h = remaining host bits; subtract 2 for network and broadcast addresses)
- Network address: AND of IP address with subnet mask
- Broadcast address: set all host bits to 1

Subnetting Example

Given: Class C network 192.168.1.0, subnet mask 255.255.255.192 (/26).

Borrowed bits $s = 2$, host bits $h = 6$.

Subnets = $2^2 = 4$. Hosts/subnet = $2^6 - 2 = 62$.

Subnets: 192.168.1.0/26, 192.168.1.64/26, 192.168.1.128/26, 192.168.1.192/26.

1.4 Routing

Routing: The process of selecting the best path for packets to travel from source to destination across an internetwork. Performed by routers using **routing tables**.

Static vs. Dynamic Routing

Parameter	Static Routing	Dynamic Routing
Configuration	Manually by admin	Automatically by routing protocols
Adaptability	Cannot adapt to changes	Adapts to topology changes
Overhead	No routing protocol overhead	Routing protocol traffic overhead
Scalability	Poor for large networks	Good; scales well
Use case	Small, stable networks	Large, dynamic networks

1.5 Unicast Routing Protocols

RIP (Routing Information Protocol)

RIP — Distance Vector Protocol

Algorithm: Bellman-Ford. Each router maintains a table of (destination, next-hop, distance) and shares it with **direct neighbours** every 30 seconds.

Metric: Hop count. **Maximum hops:** 15 (16 = infinity/unreachable).

Versions: RIPv1 (classful, broadcast), RIPv2 (classless, multicast 224.0.0.9).

Problems: Slow convergence, count-to-infinity problem, not suitable for large networks.

OSPF (Open Shortest Path First)

OSPF — Link State Protocol

Algorithm: Dijkstra's Shortest Path First. Each router floods **Link State Advertisements (LSAs)** to all routers; every router builds a complete topology map.

Metric: Cost (based on bandwidth). **No hop limit.**

Features: Fast convergence, hierarchical (areas), supports VLSM/CIDR, authentication.

Hello packets: Sent every 10 seconds to discover and maintain neighbours.

BGP (Border Gateway Protocol)

BGP — Path Vector Protocol

Use: Inter-domain routing between Autonomous Systems (AS) on the Internet. The Internet's core routing protocol.

Algorithm: Path vector; each route advertisement includes the full AS path.

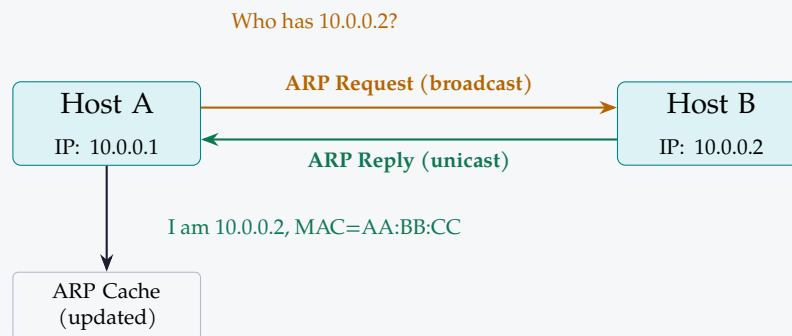
Metric: Policy-based (AS path length, local preference, MED, etc.).

Port: TCP port 179. **Types:** iBGP (within AS), eBGP (between AS).

1.6 Other Network Layer Protocols

Protocol	Function
ARP	Address Resolution Protocol. Maps IP address → MAC address. Broadcasts ARP request; target replies with its MAC.
RARP	Reverse ARP. Maps MAC address → IP address. Used by diskless workstations (replaced by DHCP).
ICMP	Internet Control Message Protocol. Reports errors and diagnostic info (e.g. ping uses ICMP Echo Request/Reply).
IP (IPv4)	Connectionless, unreliable packet delivery. 32-bit addressing, fragmentation, TTL field.
IPv6	128-bit addressing (340 undecillion addresses), simplified header, no fragmentation by routers, built-in IPSec.

ARP — Address Resolution Process



IPv4 vs. IPv6 Key Differences:

Feature	IPv4	IPv6
Address size	32 bits (4 bytes)	128 bits (16 bytes)
Notation	Dotted decimal (x.x.x.x)	Colon-hex (x:x:x:x:x:x:x:x)
Header size	20–60 bytes (variable)	40 bytes (fixed)
Fragmentation	By routers and hosts	By source host only
Checksum	Present in header	Removed (handled by upper layers)
Security	Optional (IPSec)	Mandatory (IPSec built-in)

2. Transport Layer

Transport Layer (Layer 4): Provides *process-to-process* (end-to-end) delivery of data. Handles segmentation, reassembly, flow control, error control, and multiplexing using **port numbers**.

2.1 UDP (User Datagram Protocol)

UDP — Connectionless, Unreliable

Features: No connection setup, no acknowledgement, no flow/error control. Minimal overhead.

Header: 8 bytes — Source Port (16b) | Destination Port (16b) | Length (16b) | Checksum (16b).

Use cases: DNS, DHCP, SNMP, VoIP, video streaming, online gaming.

Advantage: Very fast, low latency. **Disadvantage:** No reliability guarantee.

2.2 TCP (Transmission Control Protocol)

TCP — Connection-Oriented, Reliable

Features: Full-duplex, byte-stream, reliable delivery, flow control (sliding window), congestion control, ordered delivery.

Header: 20–60 bytes. Key fields: Source/Dest Port, Sequence No., ACK No., Flags (SYN, ACK, FIN, RST), Window Size, Checksum.

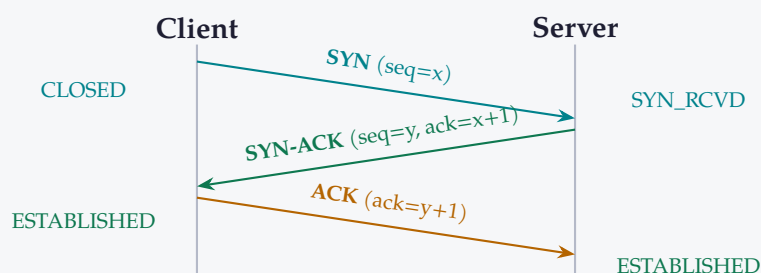
Use cases: HTTP, FTP, SMTP, SSH, Telnet.

TCP Three-Way Handshake (Connection Establishment):

1. **SYN:** Client sends SYN segment with initial sequence number x .
2. **SYN-ACK:** Server responds with SYN (y) + ACK ($x + 1$).
3. **ACK:** Client sends ACK ($y + 1$). Connection established.

TCP Connection Termination (Four-Way): FIN → ACK → FIN → ACK.

TCP Three-Way Handshake



2.3 Congestion Control

Congestion: A state where the network cannot handle the offered load; routers drop packets, causing retransmissions and further congestion. Congestion control prevents/reduces this.

Open Loop Congestion Control

Prevents congestion **before it occurs** through good design policies:

- **Retransmission policy:** Retransmit only when necessary (use timers wisely).
- **Window policy:** Use selective repeat (wastes less bandwidth than GBN).
- **Discarding policy:** Routers discard less sensitive packets first.
- **Acknowledgement policy:** Receiver delays ACK to reduce ACK traffic.
- **Admission policy:** Routers reject new connections if network is near capacity.

Closed Loop Congestion Control

Reacts to congestion **after it occurs**:

- **Backpressure:** Congested node sends a signal to upstream node to slow down. Works hop-by-hop.
- **Choke packet:** Congested router sends a choke packet directly to the source to reduce its transmission rate.
- **Implicit signalling:** Source detects congestion from dropped packets or increased delay (TCP's approach).
- **Explicit signalling:** Router sets a bit in the packet header to warn source/destination (ECN — Explicit Congestion Notification).

2.4 Quality of Service (QoS)

QoS: The ability of a network to provide different levels of service to different traffic types. Key metrics: **bandwidth**, **delay**, **jitter** (delay variation), **packet loss**.

Leaky Bucket Algorithm

Leaky Bucket Algorithm

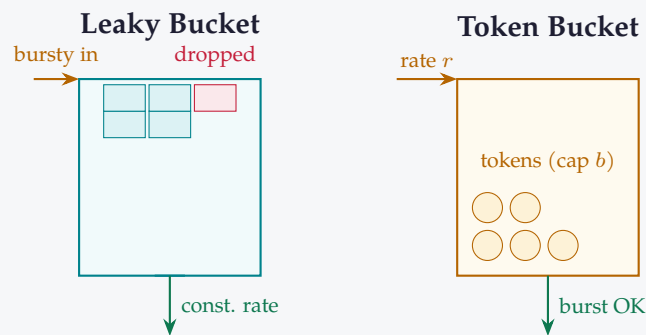
Concept: A bucket with a small hole at the bottom. Packets enter the bucket (queue) at any rate; they leave at a **constant rate** (the leak rate). Smooths bursty traffic into a steady stream.

Operation:

- If bucket is not full: incoming packet is added to the queue.
- If bucket is full: packet is **discarded** (overflow).
- Packets leave the bucket at a fixed rate r (tokens/second).

Advantage: Enforces constant output rate. **Disadvantage:** Cannot handle bursts; excess packets dropped.

Leaky Bucket vs Token Bucket



Token Bucket Algorithm

Token Bucket Algorithm

Concept: Tokens are generated at a constant rate r and stored in a bucket (capacity b). A packet can only be sent if a token is available (one token consumed per packet).

Operation:

- Tokens accumulate up to bucket capacity b .
- To send a packet: remove one token from bucket.
- If no token available: packet must wait (or be discarded).

Maximum burst size: b packets (when bucket is full).

Advantage: Allows controlled bursts up to b . More flexible than leaky bucket.

Feature	Leaky Bucket	Token Bucket
Output rate	Constant (fixed)	Variable (up to burst limit)
Burst handling	Excess packets dropped	Bursts allowed up to b tokens
Flexibility	Less flexible	More flexible
Use case	Traffic shaping	Traffic policing with burst allowance

Quick Revision — Module 3

Network Layer & Transport Layer | EC602

Key Facts at a Glance

Class A range	1–126; mask 255.0.0.0
Class B range	128–191; mask 255.255.0.0
Class C range	192–223; mask 255.255.255.0
Subnets	2^s (s = borrowed bits)
Hosts per subnet	$2^h - 2$ (h = host bits)
RIP metric / max hops	Hop count / 15
OSPF metric	Cost (bandwidth-based)
BGP use	Inter-AS routing; TCP port 179
ARP function	IP → MAC address resolution
ICMP use	Error reporting; ping (Echo Request/Reply)
IPv6 address size	128 bits
TCP handshake	SYN → SYN-ACK → ACK
UDP header size	8 bytes
TCP header size	20–60 bytes
Leaky bucket output	Constant rate
Token bucket burst	Up to b tokens

One-Line Definitions

- **Router:** Layer 3 device that routes packets between networks using IP addresses.
- **Gateway:** Protocol converter connecting networks with different protocols.
- **Subnetting:** Dividing a network into smaller subnets by borrowing host bits.
- **RIP:** Distance vector routing protocol; metric = hop count; max 15 hops.
- **OSPF:** Link state routing protocol; uses Dijkstra's algorithm; metric = cost.
- **BGP:** Path vector protocol for inter-AS routing on the Internet.
- **ARP:** Resolves IP address to MAC address via broadcast.
- **ICMP:** Reports network errors and provides diagnostic functions (ping).
- **TCP:** Reliable, connection-oriented transport protocol with flow/error control.

- **UDP:** Unreliable, connectionless transport protocol; low overhead, fast.
- **Congestion:** Network overload causing packet drops and retransmissions.
- **Leaky Bucket:** Traffic shaping algorithm enforcing constant output rate.
- **Token Bucket:** Traffic policing algorithm allowing controlled bursts.

Frequently Asked / Viva Questions

Q1. What is the difference between a switch and a router?

Switch operates at Layer 2 (MAC addresses, within a network); router operates at Layer 3 (IP addresses, between networks).

Q2. What is subnetting and why is it used?

Dividing a network into smaller subnets. Used to reduce broadcast domains, improve security, and efficient IP address utilisation.

Q3. How many hosts can a /26 subnet support?

$2^6 - 2 = 62$ usable hosts.

Q4. Difference between RIP and OSPF?

RIP: distance vector, hop count metric, max 15 hops, slow convergence. OSPF: link state, cost metric, no hop limit, fast convergence.

Q5. What is BGP used for?

Inter-domain routing between Autonomous Systems on the Internet; uses path vector algorithm.

Q6. What does ARP do?

Resolves an IP address to a MAC address by broadcasting an ARP request on the local network.

Q7. Difference between TCP and UDP?

TCP: reliable, connection-oriented, ordered, flow/error control. UDP: unreliable, connectionless, no ordering, minimal overhead.

Q8. What is the TCP three-way handshake?

SYN (client) → SYN-ACK (server) → ACK (client). Establishes a reliable connection.

Q9. What is the difference between leaky bucket and token bucket?

Leaky bucket: constant output rate, excess dropped. Token bucket: variable output up to burst size b , more flexible.

Q10. What is a choke packet?

A packet sent by a congested router directly to the source to reduce its transmission rate (closed-loop congestion control).

Frequently Asked / Viva Questions (contd.)**Q11. What is the purpose of TTL in IP header?**

Time To Live; decremented by each router. When $TTL = 0$, packet is discarded. Prevents infinite routing loops.

Q12. What are the main improvements of IPv6 over IPv4?

128-bit addressing, fixed 40-byte header, no router fragmentation, built-in IPSec, no broadcast (uses multicast).

Module 4

Application Layer Protocols

1. Application Layer Protocols

Application Layer (Layer 7): Provides network services directly to user applications. Protocols define the format and rules for communication between applications on different hosts.

Protocol	Port	Transport	Function
DNS	53	UDP/TCP	Resolves domain names to IP addresses.
SMTP	25	TCP	Sends email from client to server (or server to server).
SNMP	161	UDP	Manages and monitors network devices.
FTP	20/21	TCP	File transfer; port 21 = control, port 20 = data.
HTTP	80	TCP	Transfers web pages (HyperText Transfer Protocol).
HTTPS	443	TCP	Secure HTTP using TLS/SSL encryption.

1.1 DNS (Domain Name System)

DNS

Function: Translates human-readable domain names (e.g. `www.google.com`) to IP addresses.

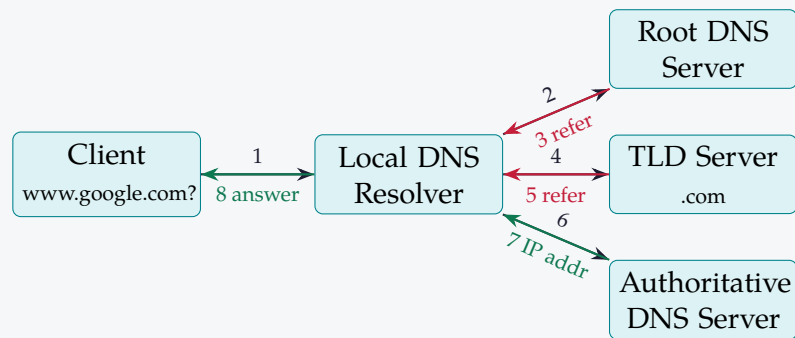
Hierarchy: Root DNS → TLD servers (.com, .org, .in) → Authoritative DNS servers.

Resolution types:

- **Recursive:** DNS resolver queries on behalf of client until answer found.
- **Iterative:** DNS server returns best referral; client queries next server itself.

Caching: DNS responses cached with TTL to reduce lookup time.

DNS Resolution Hierarchy



1.2 HTTP and WWW

HTTP (HyperText Transfer Protocol): Application layer protocol for transferring web content. Stateless, request-response model. Client sends HTTP request; server responds with status code and content.

HTTP Methods: GET (retrieve), POST (submit data), PUT (update), DELETE (remove), HEAD (headers only).

Common Status Codes: 200 OK, 301 Moved Permanently, 404 Not Found, 500 Internal Server Error.

WWW: World Wide Web — a system of interlinked hypertext documents accessed via HTTP. Uses URLs, HTML, and browsers.

2. Network Security

2.1 Cryptography

Cryptography: The science of securing information by transforming it into an unreadable format (ciphertext) using an algorithm and a key. Only authorised parties with the correct key can decrypt it.

Symmetric Key (Private Key) Cryptography

Symmetric Key Cryptography

Concept: Same key used for both encryption and decryption. Both sender and receiver must share the secret key securely.

Algorithms: DES (56-bit key), 3DES, AES (128/192/256-bit), RC4, Blowfish.

Advantages: Fast, low computational overhead, suitable for bulk data encryption.

Disadvantages: Key distribution problem — how to securely share the key? Does not scale well (n users need $n(n-1)/2$ keys).

Asymmetric Key (Public Key) Cryptography

Asymmetric Key Cryptography

Concept: Uses a **key pair** — a **public key** (shared openly) and a **private key** (kept secret). Data encrypted with public key can only be decrypted with the corresponding private key.

Algorithms: RSA, Diffie-Hellman, ECC (Elliptic Curve Cryptography).

Encryption: Sender encrypts with receiver's **public key**; receiver decrypts with **private key**.

Advantages: Solves key distribution problem; enables digital signatures.

Disadvantages: Computationally expensive; much slower than symmetric encryption.

Feature	Symmetric	Asymmetric
Keys used	Same key (shared secret)	Public + Private key pair
Speed	Fast	Slow
Key distribution	Difficult (must share secretly)	Easy (public key is open)
Scalability	Poor ($n^2/2$ keys)	Good ($2n$ keys for n users)
Examples	AES, DES, RC4	RSA, Diffie-Hellman, ECC
Use case	Bulk data encryption	Key exchange, digital signatures

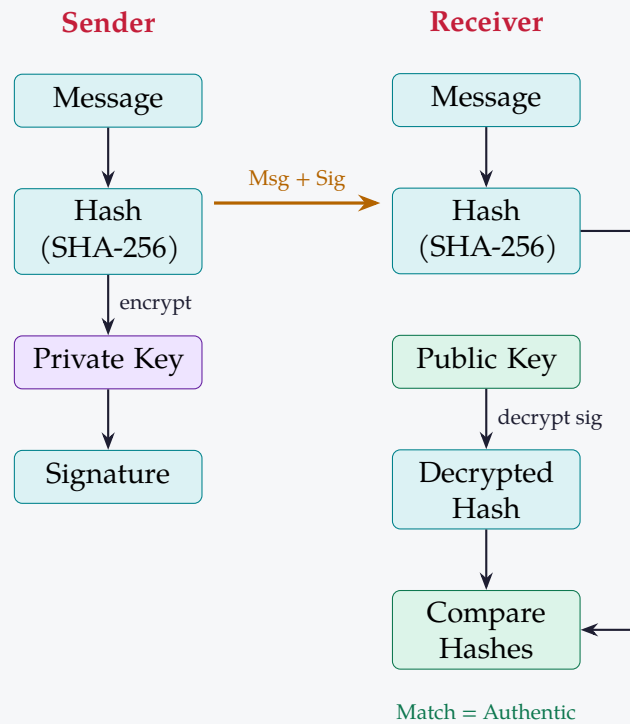
2.2 Digital Signature

Digital Signature: A cryptographic mechanism that provides **authentication**, **integrity**, and **non-repudiation**. Created by encrypting a message digest (hash) with the sender's **private key**.

Process:

1. Sender computes hash of message using a hash function (e.g. SHA-256).
2. Sender encrypts the hash with their **private key** — this is the digital signature.
3. Receiver decrypts the signature using sender's **public key** to get the hash.
4. Receiver independently computes hash of received message.
5. If both hashes match: message is authentic and unaltered.

Digital Signature — Sign & Verify



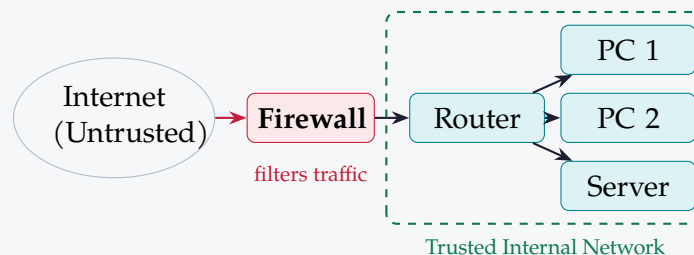
2.3 Firewalls

Firewall: A network security device (hardware or software) that monitors and controls incoming/outgoing network traffic based on predefined security rules. Acts as a barrier between trusted internal network and untrusted external network.

Types of Firewalls:

Type	Description
Packet Filter	Inspects packets at network layer; filters based on IP, port, protocol. Stateless.
Stateful Inspection	Tracks connection state; allows only packets belonging to established connections.
Application Gateway	Proxy-based; inspects application layer data (HTTP, FTP). Most secure but slowest.
Circuit-level Gateway	Monitors TCP handshake; does not inspect packet content.

Firewall Placement in a Network



3. Modern Topics

3.1 ISDN (Integrated Services Digital Network)

ISDN: A set of communication standards for simultaneous digital transmission of voice, video, data, and other services over traditional telephone copper wire. Replaced by DSL and broadband.

ISDN Channels:

- **B channel (Bearer):** 64 Kbps; carries user data (voice, video, data).
- **D channel (Delta):** 16 or 64 Kbps; carries signalling and control information.

ISDN Interfaces:

Interface	Description
BRI (Basic Rate)	$2B + D = 2 \times 64 \text{ Kbps} + 16 \text{ Kbps}$ D = 144 Kbps. For home/small office.
PRI (Primary Rate)	$23B + D$ (North America) or $30B + D$ (Europe) = 1.544 / 2.048 Mbps. For enterprises.

3.2 ATM (Asynchronous Transfer Mode)

ATM: A cell-switching and multiplexing technology that uses fixed-size **53-byte cells** (5-byte header + 48-byte payload) for transmitting voice, video, and data over high-speed networks.

Key Features:

- Fixed 53-byte cells enable predictable, low-latency switching.
- Connection-oriented: virtual circuits (VC) established before data transfer.
- Supports QoS guarantees for different traffic types (CBR, VBR, ABR, UBR).
- **VPI/VCI:** Virtual Path Identifier / Virtual Channel Identifier used for routing cells.

3.3 DSL (Digital Subscriber Line)

DSL: A family of technologies that transmit digital data over existing telephone copper wire pairs at high speeds. Uses frequencies above the voice band (above 4 kHz).

Type	Speed	Description
ADSL (Asymmetric)	Down: 8 Mbps, Up: 1 Mbps	Different download/upload speeds. Most common for home use.
SDSL (Symmetric)	2 Mbps both ways	Equal upload/download. Used by businesses.
VDSL (Very High)	Down: 52 Mbps, Up: 16 Mbps	Very high speed over short distances.

DSL Modem: Connects user's equipment to the telephone line. Separates voice (0–4 kHz) and data (above 4 kHz) using a **splitter**.

3.4 Cable Modem

Cable Modem: A device that provides Internet access over the cable TV (CATV) infrastructure using coaxial cable. Uses **DOCSIS** (Data Over Cable Service Interface Specification) standard.

Key Points:

- Shared medium: all users in a neighbourhood share the same cable segment (bandwidth shared).
- Downstream (TV to user): 6 MHz channel, up to 30–40 Mbps.
- Upstream (user to TV headend): narrower band, typically 5–42 MHz.
- **HFC (Hybrid Fibre-Coaxial):** Fibre from headend to neighbourhood node; coaxial to homes.

3.5 Wireless LAN — IEEE 802.11

IEEE 802.11 (Wi-Fi): A set of standards for wireless local area networking. Uses radio frequencies (2.4 GHz and 5 GHz bands). MAC protocol: CSMA/CA.

IEEE 802.11 Standards:

Standard	Year	Frequency	Max Speed
802.11b	1999	2.4 GHz	11 Mbps
802.11a	1999	5 GHz	54 Mbps
802.11g	2003	2.4 GHz	54 Mbps
802.11n	2009	2.4/5 GHz	600 Mbps (MIMO)
802.11ac	2013	5 GHz	1+ Gbps

Architecture:

- **Infrastructure mode:** Devices connect through an **Access Point (AP)**. AP connects to wired network. Most common.
- **Ad-hoc mode:** Devices communicate directly without an AP (peer-to-peer).
- **BSS (Basic Service Set):** Single AP + associated stations.
- **ESS (Extended Service Set):** Multiple BSSs connected via a distribution system.

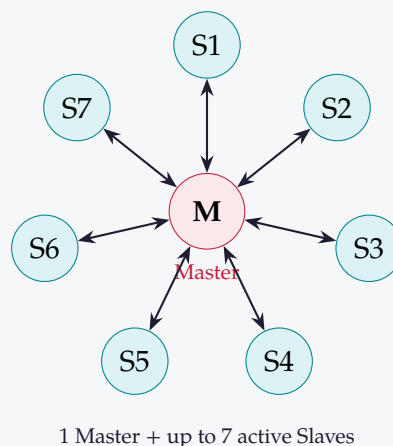
Security: WEP (weak, deprecated) → WPA → WPA2 (AES-based, current standard) → WPA3.

3.6 Bluetooth

Bluetooth (IEEE 802.15.1): A short-range wireless technology for personal area networks (PAN). Operates at 2.4 GHz ISM band. Range: ~10 m (Class 2). Uses **FHSS** (Frequency Hopping Spread Spectrum) — hops 1600 times/second across 79 channels.

Bluetooth Network (Piconet):

- **Piconet:** One **master** + up to 7 active **slaves**. Master controls timing and frequency hopping.
- **Scatternet:** Multiple overlapping piconets; a device can be slave in multiple piconets or master in one and slave in another.

Bluetooth Piconet Structure**Bluetooth Versions:**

Version	Key Feature
1.2	Basic rate 1 Mbps; AFH (Adaptive Frequency Hopping)
2.0 + EDR	Enhanced Data Rate; up to 3 Mbps
4.0 (BLE)	Bluetooth Low Energy; very low power; IoT devices
5.0	2× speed, 4× range, 8× broadcast capacity vs. 4.2

Quick Revision — Module 4

Application Layer, Security & Modern Topics | EC602

Key Facts at a Glance

DNS port	53 (UDP/TCP)
SMTP port	25 (TCP)
FTP ports	21 (control), 20 (data)
HTTP / HTTPS ports	80 / 443
SNMP port	161 (UDP)
Symmetric key	Same key for encrypt/decrypt (AES, DES)
Asymmetric key	Public key encrypts; private key decrypts (RSA)
Digital signature	Hash encrypted with sender's private key
ISDN BRI	$2B + D = 144$ Kbps
ISDN PRI (Europe)	$30B + D = 2.048$ Mbps
ATM cell size	53 bytes (5 header + 48 payload)
ADSL speed	Down 8 Mbps / Up 1 Mbps
802.11b speed	11 Mbps @ 2.4 GHz
802.11g/a speed	54 Mbps
Bluetooth range	~10 m; 2.4 GHz; FHSS; 1600 hops/sec
Piconet	1 master + up to 7 active slaves

One-Line Definitions

- **DNS:** Translates domain names to IP addresses using a hierarchical distributed database.
- **HTTP:** Stateless request-response protocol for transferring web content.
- **SMTP:** Protocol for sending email between mail servers.
- **FTP:** Protocol for transferring files; uses separate control (21) and data (20) connections.
- **Symmetric Cryptography:** Same secret key used for both encryption and decryption.
- **Asymmetric Cryptography:** Public key encrypts; private key decrypts (or vice versa for signatures).
- **Digital Signature:** Hash of message encrypted with sender's private key; proves

authenticity.

- **Firewall:** Security device that filters network traffic based on rules.
- **ISDN:** Digital telephone network supporting voice, video, and data simultaneously.
- **ATM:** Cell-switching technology using fixed 53-byte cells; supports QoS.
- **DSL:** High-speed Internet over telephone copper wire using frequencies above voice band.
- **IEEE 802.11:** Wi-Fi standard; uses CSMA/CA; operates at 2.4/5 GHz.
- **Bluetooth:** Short-range PAN technology; 2.4 GHz; FHSS; piconet architecture.

Frequently Asked / Viva Questions

Q1. What is DNS and how does it work?

DNS translates domain names to IP addresses. Client queries local DNS resolver; if not cached, resolver queries root → TLD → authoritative DNS servers recursively or iteratively.

Q2. What is the difference between symmetric and asymmetric cryptography?

Symmetric: same key for encrypt/decrypt, fast, key distribution problem.
Asymmetric: public/private key pair, slower, solves key distribution.

Q3. How does a digital signature work?

Sender hashes the message and encrypts the hash with their private key. Receiver decrypts with sender's public key and compares hashes to verify authenticity and integrity.

Q4. What is a firewall?

A security device that monitors and filters network traffic based on predefined rules, acting as a barrier between trusted and untrusted networks.

Q5. What is the difference between packet filter and stateful firewall?

Packet filter: stateless, inspects each packet independently. Stateful: tracks connection state, allows only packets belonging to established connections.

Q6. What is ATM and what is its cell size?

Asynchronous Transfer Mode; cell-switching technology using fixed 53-byte cells (5-byte header + 48-byte payload).

Q7. What is ADSL?

Asymmetric DSL; provides higher download speed (up to 8 Mbps) than upload speed (up to 1 Mbps) over telephone copper wire.

Q8. What MAC protocol does IEEE 802.11 use?

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).

Frequently Asked / Viva Questions (contd.)**Q9. What is a piconet in Bluetooth?**

A small network of one master and up to 7 active slave devices communicating on the same frequency hopping sequence.

Q10. What is FHSS in Bluetooth?

Frequency Hopping Spread Spectrum; Bluetooth hops across 79 channels 1600 times per second to reduce interference and improve security.

Q11. What is the difference between infrastructure and ad-hoc Wi-Fi mode?

Infrastructure: devices connect through an Access Point. Ad-hoc: devices communicate directly without an AP.

Q12. What is ISDN BRI?

Basic Rate Interface: $2B + D$ channels = $2 \times 64 \text{ Kbps} + 16 \text{ Kbps} = 144 \text{ Kbps}$ total. Used for home/small office ISDN connections.

Application Layer Protocols Summary

Protocol	Port	Transport	Purpose
DNS	53	UDP/TCP	Name resolution
SMTP	25	TCP	Send email
FTP	20/21	TCP	File transfer
HTTP	80	TCP	Web pages
HTTPS	443	TCP	Secure web
SNMP	161	UDP	Network management